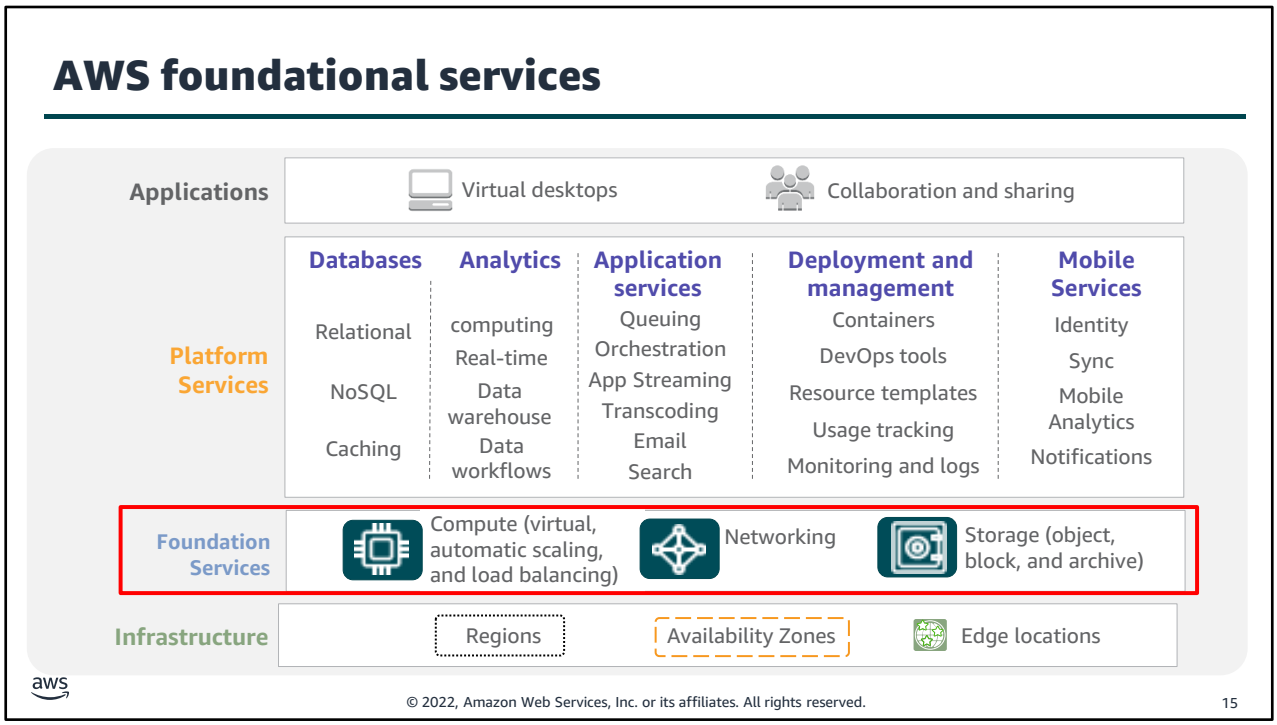

As discussed previously, the AWS Global Infrastructure can be broken down into three elements: Regions, Availability Zones, and Points of Presence, which include edge locations. This infrastructure provides the platform for a broad set of services, such as networking, storage, compute services, and databases—and these services are delivered as an on-demand utility that is available in seconds, with pay-as-you-go pricing.

For accessibility: Marketing diagram showing infrastructure at the bottom, consisting of Regions, Availability Zones, and edge locations. The next level up is labeled Foundational Services and includes graphics for compute, networking, and storage. That level is highlighted. Next level up is platform services that includes databases, analytics, app services, deployment and management, and mobile services. Top layer is labeled applications and includes virtual desktops and collaboration and sharing. **End of accessibility description.**

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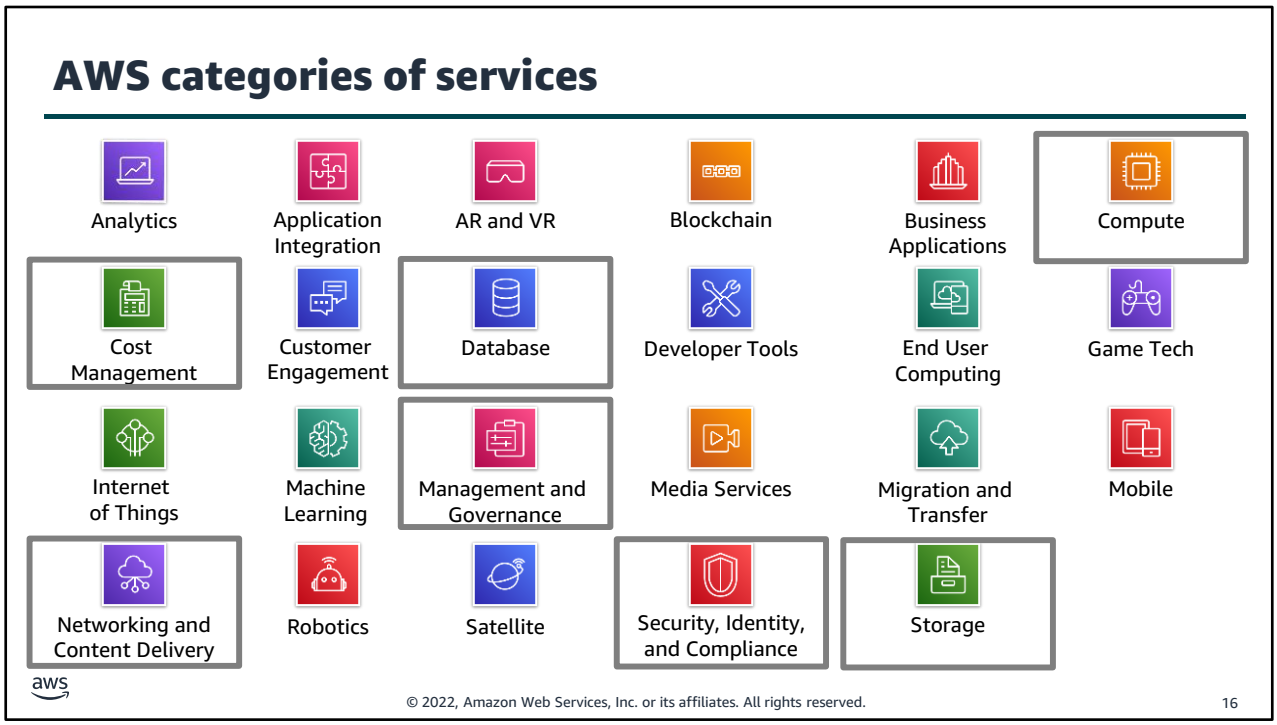

AWS offers a broad set of cloud-based services. There are 23 different product or service categories, and each category consists of one or more services. This course will not attempt to introduce you to each service. Rather, the focus of this course is on the services that are most widely used and offer the best introduction to the AWS Cloud. This course also focuses on services that are more likely to be covered in the AWS Certified Cloud Practitioner exam.

The categories that this course will discuss are highlighted on the slide: Compute, Cost Management, Database, Management and Governance, Networking and Content Delivery, Security, Identity, and Compliance, and Storage.

To learn more about AWS products, see Cloud Products at https://aws.amazon.com/products/. All AWS products are organized into the service categories that are shown here. For example, if you click **Compute,** you will see that Amazon Elastic Compute Cloud (Amazon EC2) is first on the list. The compute category also lists many other products and services.

If you click **Amazon EC2,** it takes you to the Amazon EC2 page. Each product page provides a detailed description of the product and lists some of its benefits.

Explore the different service groups to understand the categories and services within them. Now that you know how to locate information about different services, this module will discuss the highlighted service categories. **The next seven slides list the individual services —within each of the categories highlighted above—that this course will discuss.**

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Storage service category

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AWS storage services

- Amazon Simple Storage Service (Amazon S3)
- Amazon Elastic Block Store (Amazon EBS)
- Amazon Elastic File System (Amazon EFS)
- Amazon Simple Storage Service Glacier

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AWS storage services include the services listed here, and many others.

Amazon Simple Storage Service (Amazon S3) is an object storage service that offers scalability, data availability, security, and performance. Use it to store and protect any amount of data for websites, mobile apps, backup and restore, archive, enterprise applications, Internet of Things (IoT) devices, and big data analytics.

Amazon Elastic Block Store (Amazon EBS) is high-performance block storage that is designed for use with Amazon EC2 for both throughput and transaction intensive workloads. It is used for a broad range of workloads, such as relational and non-relational databases, enterprise applications, containerized applications, big data analytics engines, file systems, and media workflows.

Amazon Elastic File System (Amazon EFS) provides a scalable, fully managed elastic Network File System (NFS) file system for use with AWS Cloud services and on-premises resources. It is built to scale on demand to petabytes, growing and shrinking automatically as you add and remove files. It reduces the need to provision and manage capacity to accommodate growth.

Amazon Simple Storage Service Glacier is a secure, durable, and extremely low-cost Amazon S3 cloud storage class for data archiving and long-term backup. It is designed to deliver 11 9s of durability, and to provide comprehensive security and compliance capabilities to meet stringent regulatory requirements.

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Compute service category

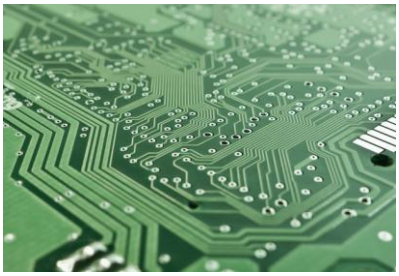

Photo from https://www.pexels.com/photo/technology-computer-lines-board-50711/

AWS Compute services

- Amazon EC2
- Amazon EC2 Auto Scaling
- Amazon Elastic Container Service (Amazon ECS)
- Amazon EC2 Container Registry
- AWS Elastic Beanstalk
- AWS Lambda
- Amazon Elastic Kubernetes Service (Amazon EKS)
- AWS Fargate

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AWS compute services include the services listed here, and many others.

Amazon Elastic Compute Cloud (Amazon EC2) provides resizable compute capacity as virtual machines in the cloud.

Amazon EC2 Auto Scaling enables you to automatically add or remove EC2 instances according to conditions that you define.

Amazon Elastic Container Service (Amazon ECS) is a highly scalable, high-performance container orchestration service that supports Docker containers.

Amazon Elastic Container Registry (Amazon ECR) is a fully-managed Docker container registry that makes it easy for developers to store, manage, and deploy Docker container images.

AWS Elastic Beanstalk is a service for deploying and scaling web applications and services on familiar servers such as Apache and Microsoft Internet Information Services (IIS).

AWS Lambda enables you to run code without provisioning or managing servers. You pay only for the compute time that you consume. There is no charge when your code is not running.

Amazon Elastic Kubernetes Service (Amazon EKS) makes it easy to deploy, manage, and scale containerized applications that use Kubernetes on AWS.

AWS Fargate is a compute engine for Amazon ECS that allows you to run containers without having to manage servers or clusters.

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Database service category

Photo from https://aws.amazon.com/compliance/data-center/data-centers/

AWS Database services

Amazon Relational Database Service

Amazon Aurora

Amazon Redshift

Amazon DynamoDB

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AWS database services include the services listed here, and many others.

Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate, and scale a relational database in the cloud. It provides resizable capacity while automating time-consuming administration tasks such as hardware provisioning, database setup, patching, and backups.

Amazon Aurora is a MySQL and PostgreSQL-compatible relational database. It is up to five times faster than standard MySQL databases and three times faster than standard PostgreSQL databases.

Amazon Redshift enables you to run analytic queries against petabytes of data that is stored locally in Amazon Redshift, and directly against exabytes of data that are stored in Amazon S3. It delivers fast performance at any scale.

Amazon DynamoDB is a key-value and document database that delivers single-digit millisecond performance at any scale, with built-in security, backup and restore, and in-memory caching.

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Networking and content delivery service category

Photo by Umberto on Unsplash

AWS networking and content delivery services

- Amazon VPC
- Elastic Load Balancing
- Amazon CloudFront
- AWS Transit Gateway
- Amazon Route 53
- AWS Direct Connect
- AWS VPN

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AWS networking and content delivery services include the services listed here, and many others.

Amazon Virtual Private Cloud (Amazon VPC) enables you to provision logically isolated sections of the AWS Cloud.

Elastic Load Balancing automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances, containers, IP addresses, and Lambda functions.

Amazon CloudFront is a fast content delivery network (CDN) service that securely delivers data, videos, applications, and application programming interfaces (APIs) to customers globally, with low latency and high transfer speeds.

AWS Transit Gateway is a service that enables customers to connect their Amazon Virtual Private Clouds (VPCs) and their on-premises networks to a single gateway.

Amazon Route 53 is a scalable cloud Domain Name System (DNS) web service designed to give you a reliable way to route end users to internet applications. It translates names (like www.example.com) into the numeric IP addresses (like 192.0.2.1) that computers use to connect to each other.

AWS Direct Connect provides a way to establish a dedicated private network connection from your data center or office to AWS, which can reduce network costs and increase bandwidth throughput.

AWS VPN provides a secure private tunnel from your network or device to the AWS global network.

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Security, identity, and compliance service category

Photo by Paweł Czerwiński on Unsplash

AWS security, identity, and compliance services

- AWS Identity and Access Management (IAM)
- AWS Organizations
- Amazon Cognito
- AWS Artifact
- AWS Key Management Service
- AWS Shield

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AWS security, identity, and compliance services include the services listed here, and many others.

AWS Identity and Access Management (IAM) enables you to manage access to AWS services and resources securely. By using IAM, you can create and manage AWS users and groups. You can use IAM permissions to allow and deny user and group access to AWS resources.

AWS Organizations allows you to restrict what services and actions are allowed in your accounts.

Amazon Cognito lets you add user sign-up, sign-in, and access control to your web and mobile apps.

AWS Artifact provides on-demand access to AWS security and compliance reports and select online agreements.

AWS Key Management Service (AWS KMS) enables you to create and manage keys. You can use AWS KMS to control the use of encryption across a wide range of AWS services and in your applications.

AWS Shield is a managed Distributed Denial of Service (DDoS) protection service that safeguards applications running on AWS.

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AWS cost management service category

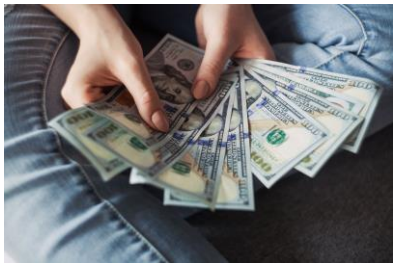

Photo by Alexander Mils on Unsplash

AWS cost management services

- AWS Cost and Usage Report
- AWS Budgets
- AWS Cost Explorer

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AWS cost management services include the services listed here, and others.

The **AWS Cost and Usage Report** contains the most comprehensive set of AWS cost and usage data available, including additional metadata about AWS services, pricing, and reservations.

AWS Budgets enables you to set custom budgets that alert you when your costs or usage exceed (or are forecasted to exceed) your budgeted amount.

AWS Cost Explorer has an easy-to-use interface that enables you to visualize, understand, and manage your AWS costs and usage over time.

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Management and governance service category

Photo by Marta Branco from Pexels

AWS management and governance services

- AWS Management Console
- AWS Config
- Amazon CloudWatch
- AWS Auto Scaling
- AWS Command Line Interface
- AWS Trusted Advisor
- AWS Well-Architected Tool
- AWS CloudTrail

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AWS management and governance services include the services listed here, and others.

The **AWS Management Console** provides a web-based user interface for accessing your AWS account.

AWS Config provides a service that helps you track resource inventory and changes.

Amazon CloudWatch allows you to monitor resources and applications.

AWS Auto Scaling provides features that allow you to scale multiple resources to meet demand.

AWS Command Line Interface provides a unified tool to manage AWS services.

AWS Trusted Advisor helps you optimize performance and security.

AWS Well-Architected Tool provides help in reviewing and improving your workloads.

AWS CloudTrail tracks user activity and API usage.

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Activity: AWS Management Console clickthrough

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In this educator-led activity, you will be asked to log in to the AWS Management Console. The activity instructions are on the next slide. You will be challenged to answer five questions. The educator will lead the class in a discussion of each question, and reveal the correct answers.

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Hands-on activity: AWS Management Console clickthrough

1. Launch the Sandbox hands-on environment and connect to the AWS Management Console.
2. Explore the AWS Management Console.
   A. Click the **Services** menu.
   B. Notice how services are grouped into service categories. For example, the **EC2** service appears in the **Compute** service category.
      Question #1: Under which service category does the **IAM** service appear?
      Question #2: Under which service category does the **Amazon VPC** service appear?
   C. Click the **Amazon VPC** service. Notice that the dropdown menu in the top-right corner displays an AWS Region (for example, it might display *N. Virginia*).
   D. Click the Region menu and switch to a different Region. For example, choose **EU (London)**.
   E. Click **Subnets** (on the left side of the screen). The Region has three subnets in it. Click the box next to one of the subnets. Notice that the bottom half of the screen now displays details about this subnet.
      Question #3: Does the subnet you selected exist at the level of the Region or at the level of the Availability Zone?
   F. Click **Your VPCs**. An existing VPC is already selected.
      Question #4: Does the VPC exist at the level of the Region or the level of the Availability Zone?
      Question #5: Which services are global instead of Regional? Check Amazon EC2, IAM, Lambda, and Route 53.

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The purpose of this activity is to expose you to the AWS Management Console. You will gain experience navigating between AWS service consoles (such as the Amazon VPC console). You will also practice navigating to services in different service categories. Finally, the console will help you distinguish whether a given service or service resource is global or Regional.

Follow the instructions on the slide. After most or all students have completed the steps document above, the educator will review the questions and answers with the whole class.

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Activity answer key

- Question #1: Under which service category does the **IAM** service appear?
  - **Answer**: **Security, Identity, & Compliance**.
- Question #2: Under which service category does the **Amazon VPC** service appear?
  - **Answer**: **Networking & Content Delivery**
- Question #3: Does the subnet that you selected exist at the level of the Region or the level of the Availability Zone?
  - **Answer**: Subnets exist at the **level of the Availability Zone.**
- Question #4: Does the VPC exist at the level of the Region or the level of the Availability Zone?
  - **Answer**: VPCs exist at the **Region level.**
- Question #5: Which of the following services are global instead of Regional? Check Amazon EC2, IAM, Lambda, and Route 53.
  - **Answer: IAM and Route 53 are global**. Amazon EC2 and Lambda are Regional.

aws

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This slide provides an answer key to the questions that were asked in the activity on the previous slide. The educator will use this slide to lead a discussion and debrief the hands-on activity.

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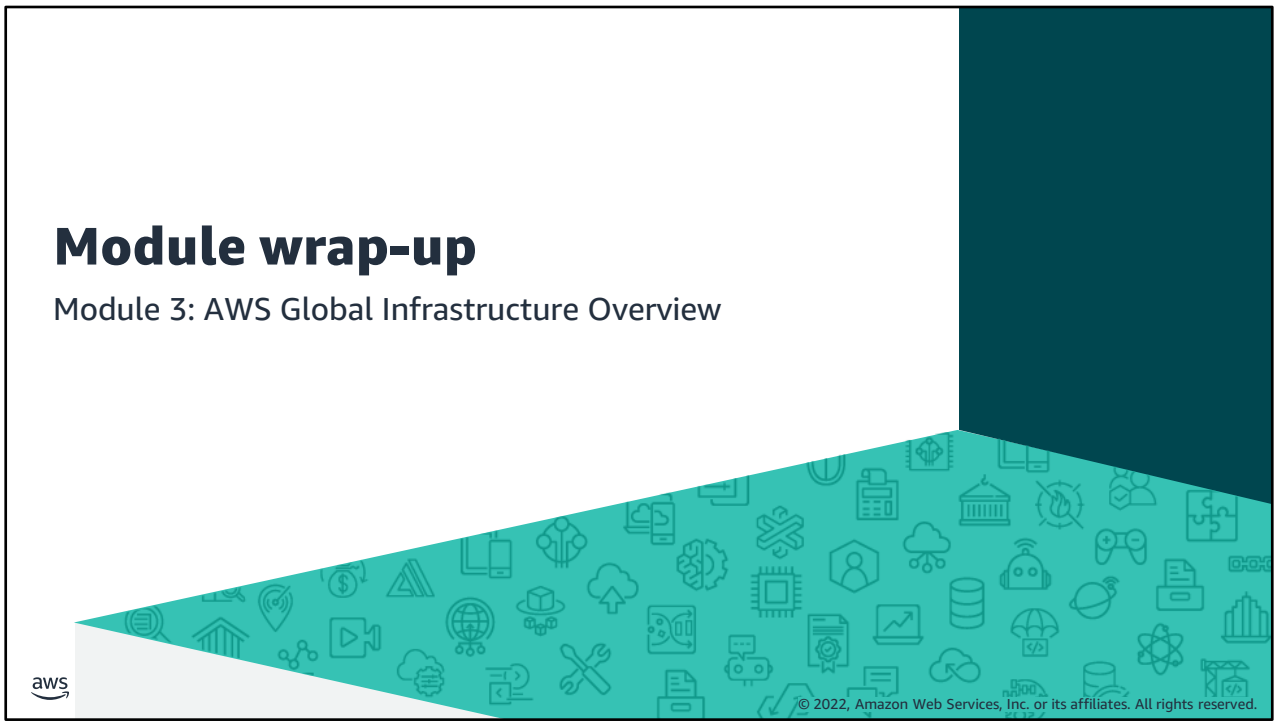

It’s now time to review the module and wrap up with a knowledge check and discussion of a practice certification exam question.

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Module summary

In summary, in this module you learned how to:

- Identify the difference between AWS Regions, Availability Zones, and edge locations
- Identify AWS service and service categories

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In summary, in this module you learned how to:
- Identify the difference between AWS Regions, Availability Zones, and edge locations
- Identify AWS service and service categories

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Complete the knowledge check

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It is now time to complete the knowledge check for this module.

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Sample exam question

Which component of AWS global infrastructure does Amazon CloudFront use to ensure low-latency delivery?

Choice	Response
A	AWS Regions
B	AWS edge locations
C	AWS Availability Zones
D	Amazon Virtual Private Cloud (Amazon VPC)

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Look at the answer choices and rule them out based on the keywords.

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Sample exam question answer

Which component of AWS global infrastructure does Amazon CloudFront use to ensure low-latency delivery?

The correct answer is B.

The keywords in the question are component of AWS global infrastructure, CloudFront, low-latency.

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The following are the keywords to recognize: component of AWS global infrastructure, CloudFront, low-latency.

The correct answer is B

Incorrect answers:
Answer A:
Answer C
Answer D

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Additional resources

- AWS Global Infrastructure: https://aws.amazon.com/about-aws/global-infrastructure/
- AWS Regional Services List: https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services/
- AWS Cloud Products: https://aws.amazon.com/products/

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The following resources provide more detail on the topics discussed in this module:

- AWS Global Infrastructure: https://aws.amazon.com/about-aws/global-infrastructure/
- AWS Regional Services List: https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services/
- AWS Cloud Products: https://aws.amazon.com/products/

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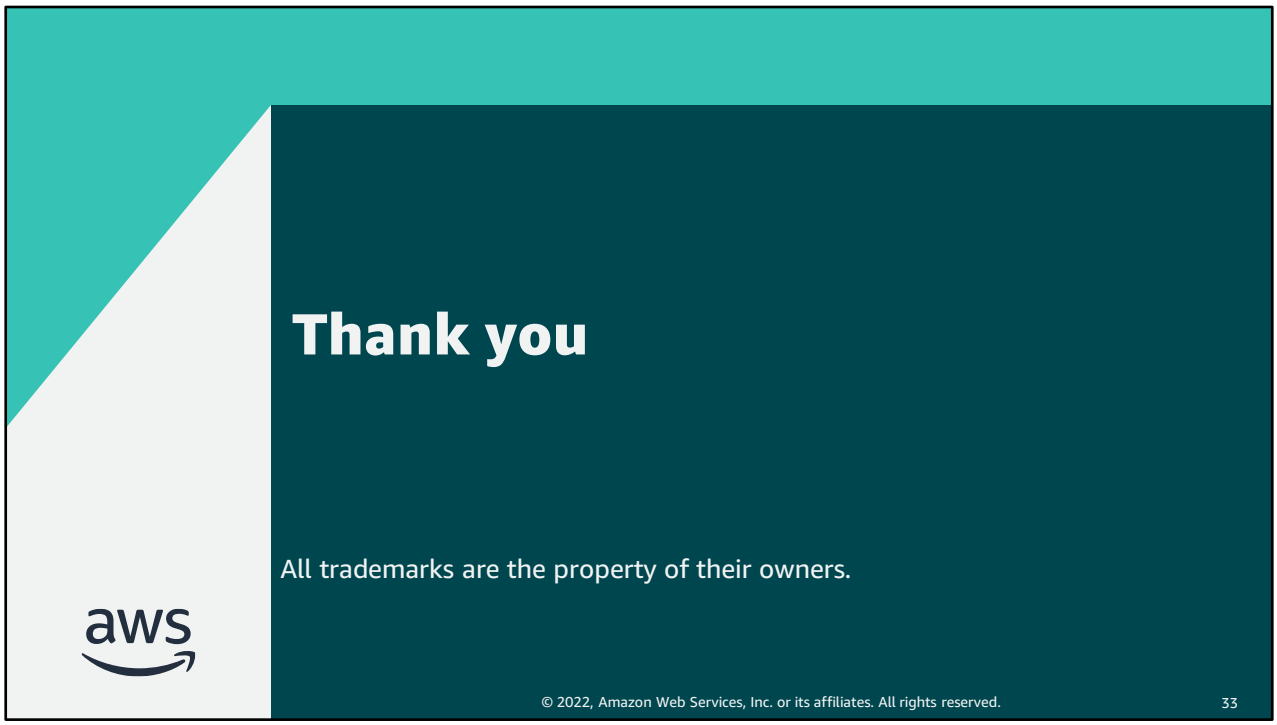

Thank you for completing this module.

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